We Broke Free From Cloud-Managed Kubernetes: Now What?

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About Us

Per Hedegaard Christiansen

- Live in Copenhagen – Denmark
- Love to travel the world
- Joined Saxo Bank 2019 and currently serves as Head of Container Platform Engineering – Previous Chief Container Platform Engineer and various other positions
- Chairman of Cloud Native Denmark – Organization behind KCD Denmark, now CND (Cloud Native Denmark)

  Last three years conference raised over 110.000 EUR for charity that foster new young talents into IT

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About Us

Kārlis Akots Gribulis

- Joined Saxo Bank in 2024 as Senior Container Platform Engineer
- Previously built Kubernetes clusters for VELUX, Accenture
- Golden Kubestronaut
- Born and raised in Rīga, Latvia. Currently living in Copenhagen, Denmark

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About Saxo Bank

- Founded 1992 in Copenhagen, Denmark
- +100 USD billion in client assets
- +1,5 mio clients
- +71.000 financial instruments
- Designated Systemically Important Financial Institution (SIFI)

- While not all core banking is on Kubernetes, or even Linux, our platform still hosts many critical and client facing workloads
- This is not an advertisement for using or avoiding either cloud or self managed Kubernetes
- Migration is not finished yet, but is well under way

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Migration Motivations

- Compliance – Exit plan
- Reliability
- Cost

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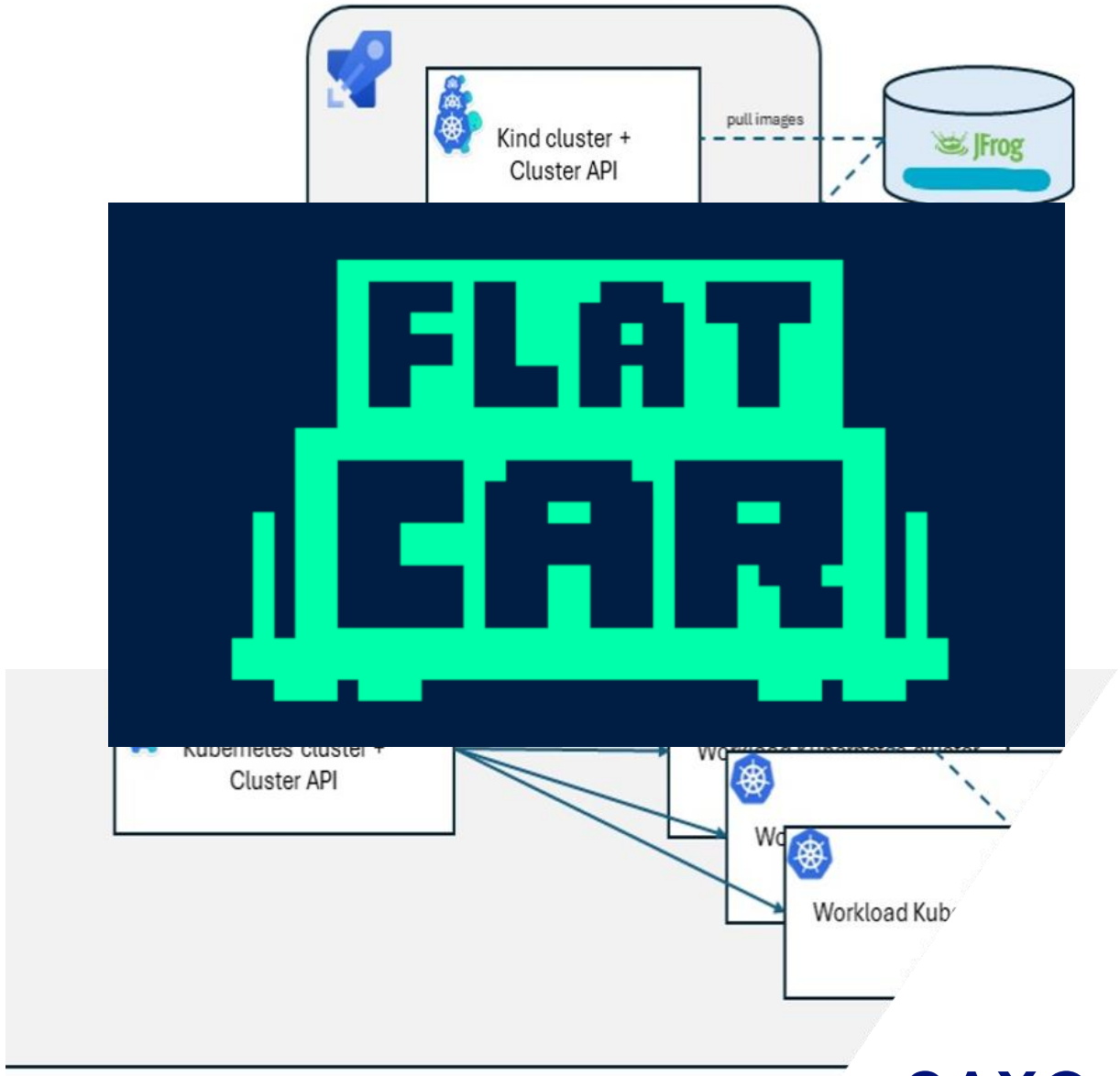
Platform Architecture

- VMware for virtualization layer
- Flatcar for OS
- Kubeadm + Cluster API for cluster lifecycle management



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Did Things *Just* Work?

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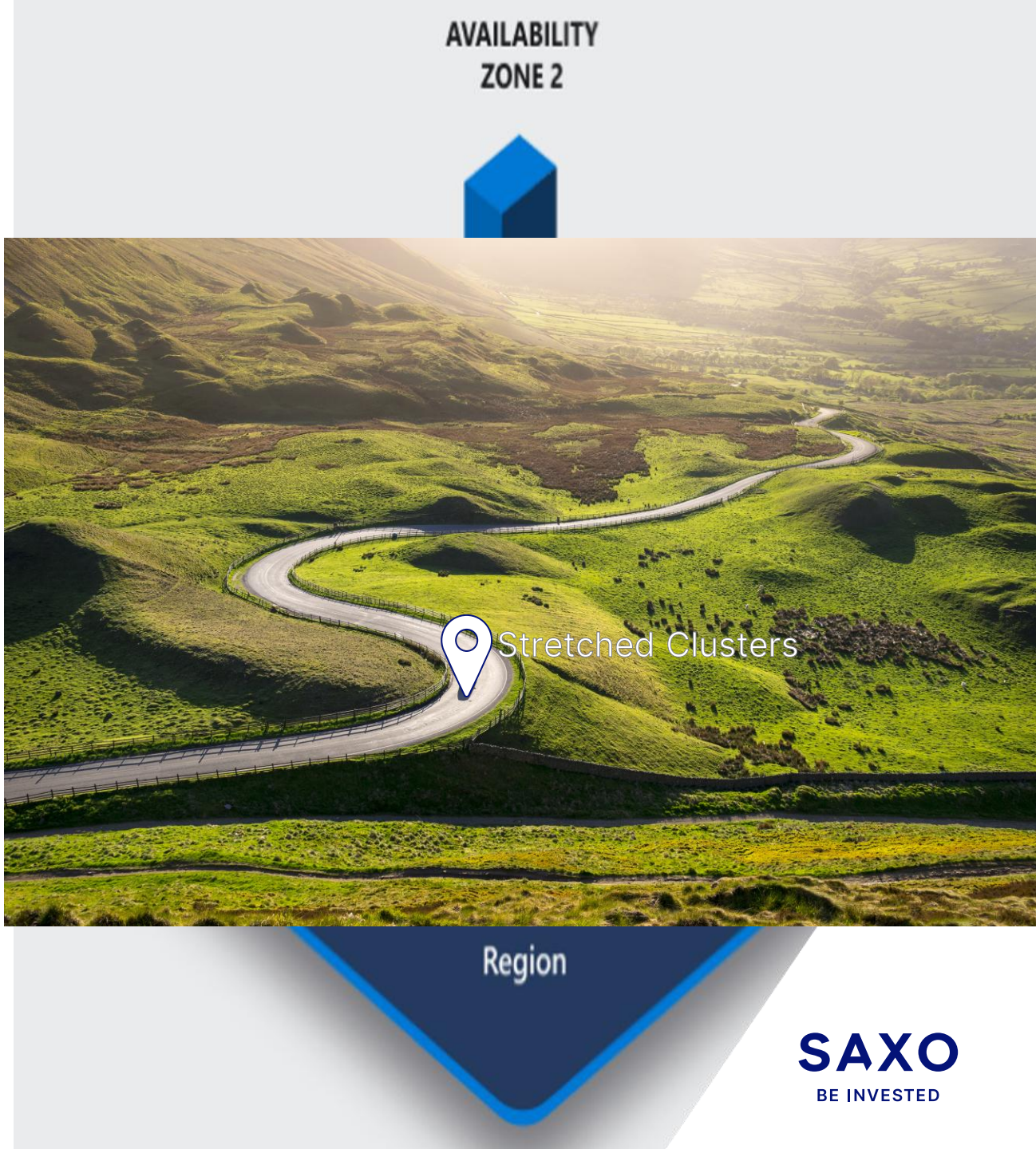
Did Things *Just* Work?

Challenges – Stretched Clusters

- Clusters stretching multiple Datacenters in Cloud is easy
- Usually Kubernetes clusters in on-premises is bound to single factory/datacenter



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Did Things *Just* Work?

Challenges - Network

- Asymmetric Routing
- No floating IPs between DCs
- Making sure traffic isn't routed to a DC that is down
  - Monthly continuity tests

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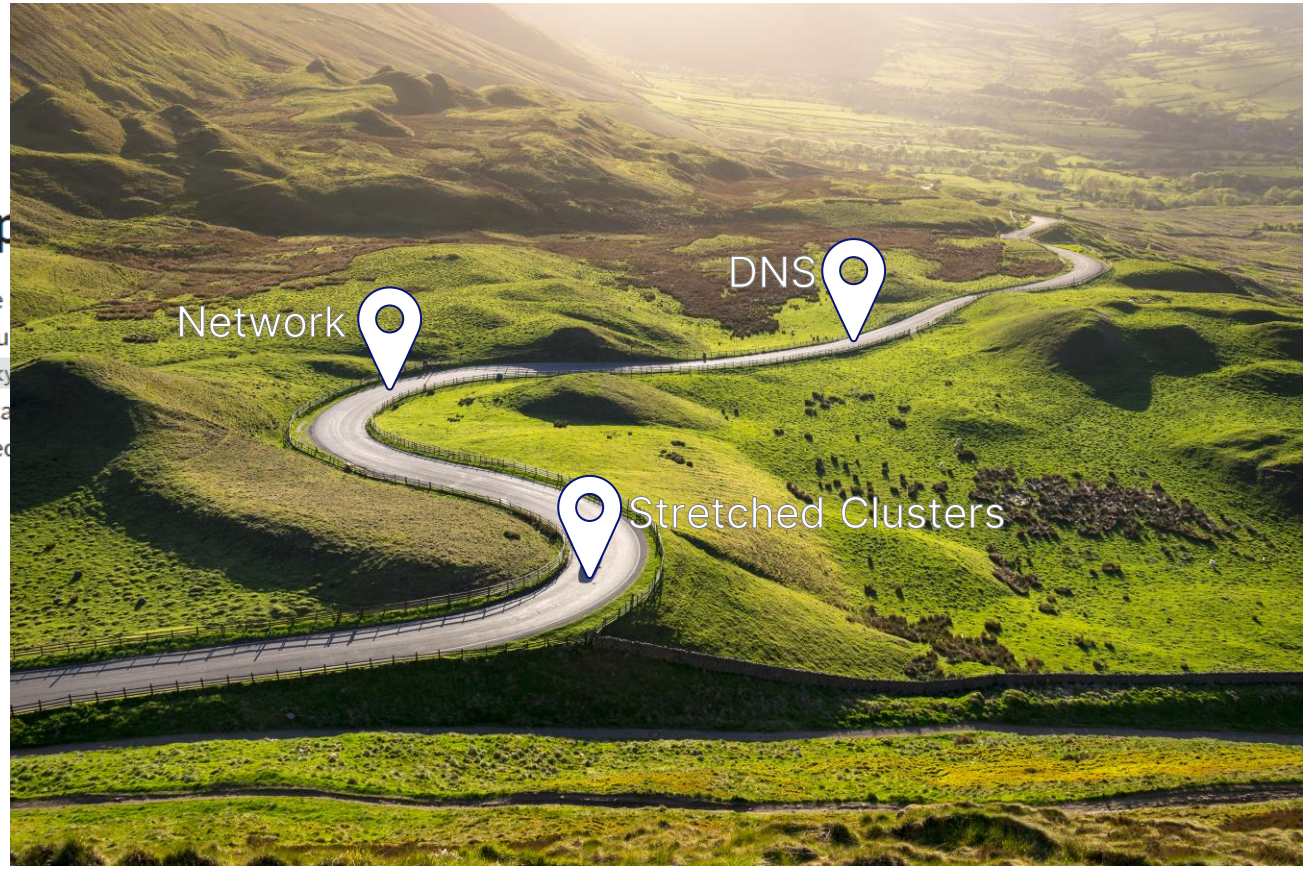
Did Things *Just* Work?

Challenges - DNS

- Re-use ExternalDNS
- AzureDNS -> CoreDNS
  - with etcd plugin (SkyDNS)

Issues:

- Recursive lookup
- No wildcard record support
- Upsert only
- No High Availability



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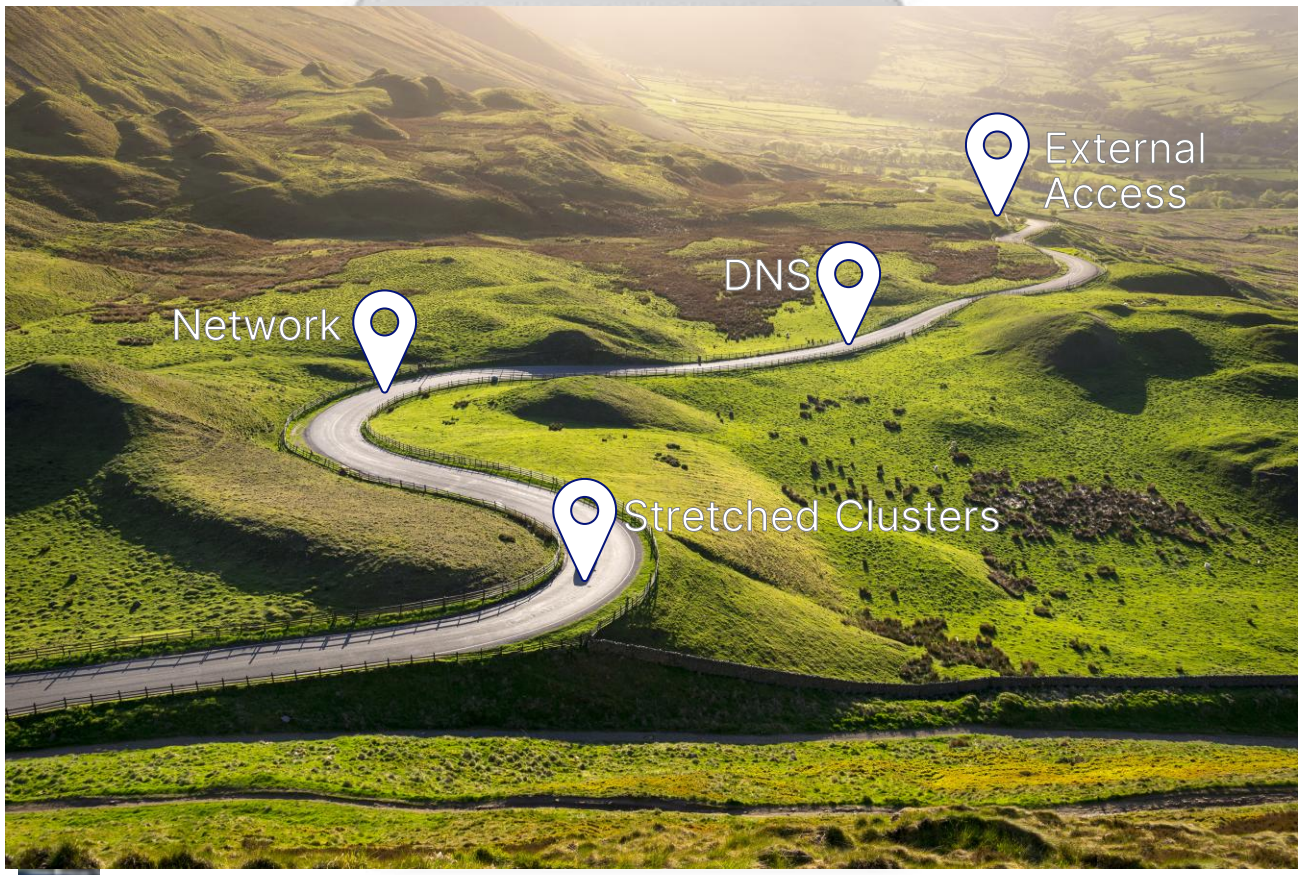
Did Things *Just* Work?

Challenges - External Access

- Using existing setup with Network appliance
- Cilium BGP for external access
  - Built a custom controller to make sure BGP speakers land on desired nodes



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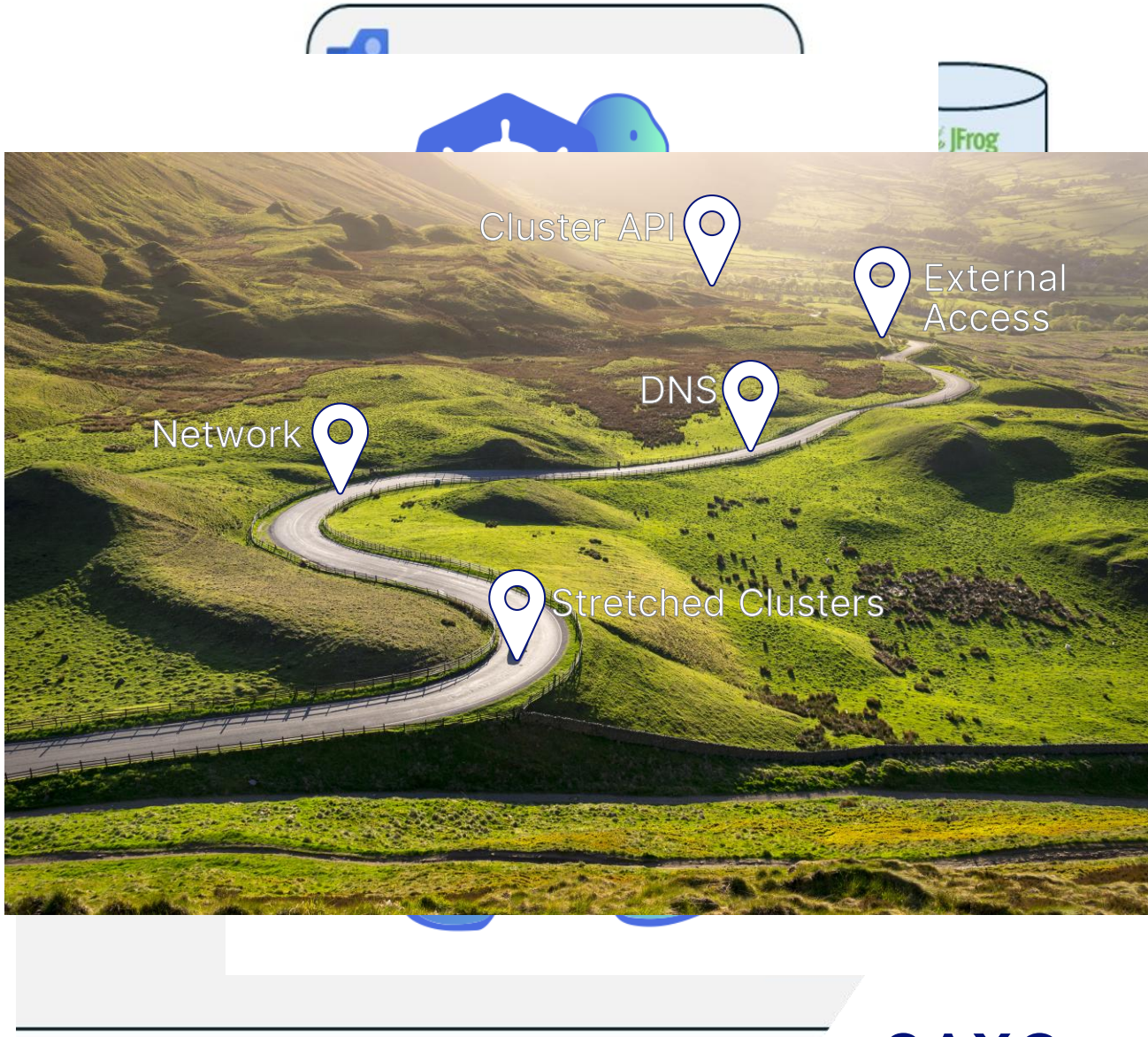
Did Things *Just* Work?

Challenges – Using CAPI for vSphere

- Needed to make our own Helm Charts
- Image builder with air gapped environment
- Many fields are immutable and feel clunky to operate
- Used Helm hooks and Kyverno to manage resource lifecycle
- Managing the original cluster



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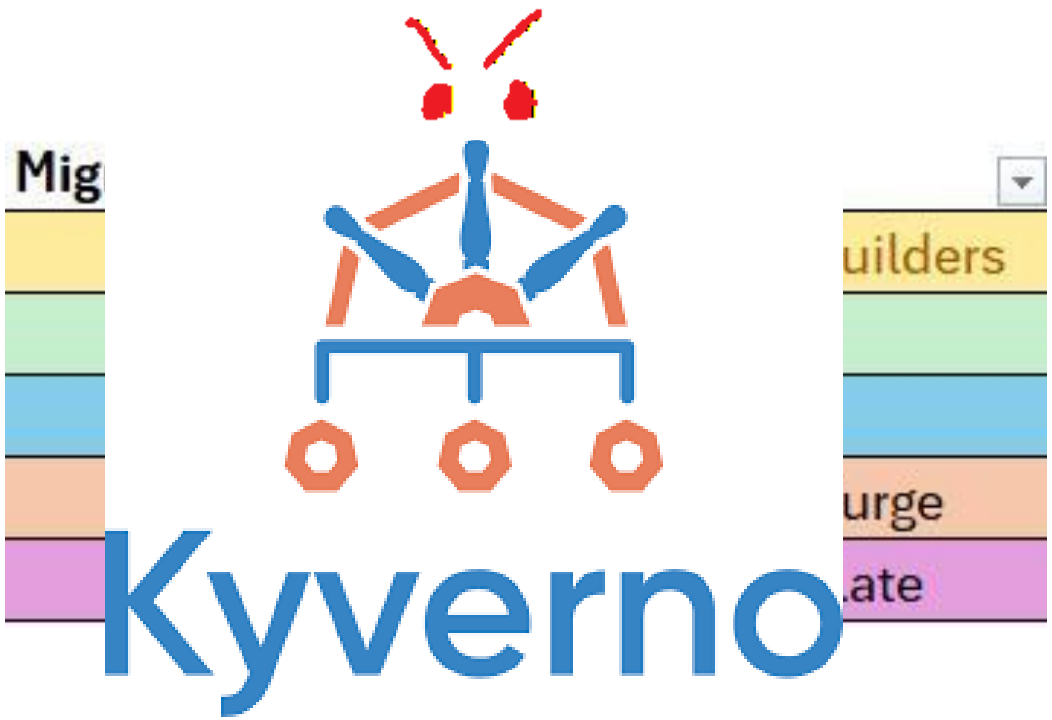
Workload Migration

Preparation & Execution

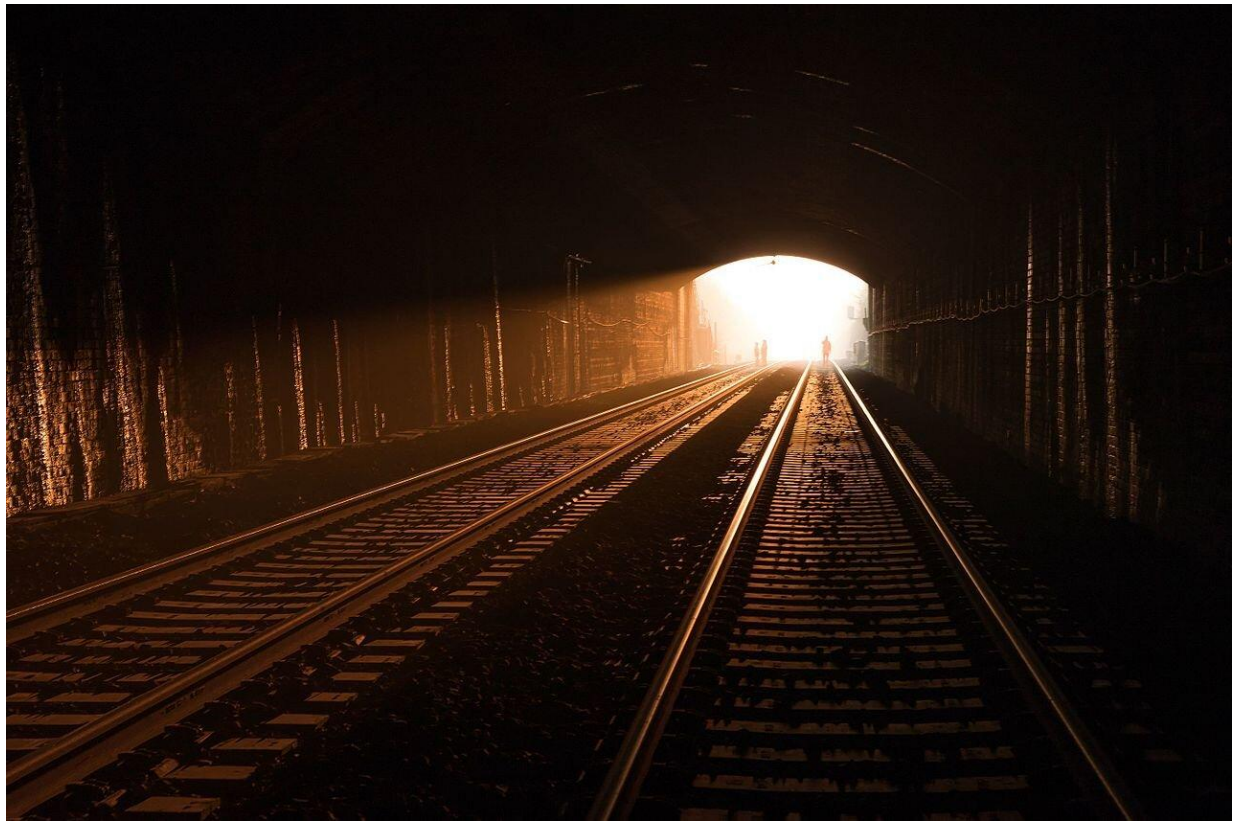
- Migration is split in multiple waves – platform, pioneers, critical workloads, the rest
- Abstracting away platform changes with Kyverno
- After all the prep, moving applications is the easiest part. Those do just work! *...mostly*



Workload Migration

Current Status

- Migration of workloads are at 10-15%
- Infrastructure optimization



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Lessons Learned

- Changing Virtualization Layer + OS
- Capacity Planning
- Know your network
- Storage

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How Much Could We Potentially Save

1. It depends on the setup that you are running and the return on investment (ROI) (3, 5 or 7 years?)

2. I Invite you to make a calculation on your running cost and see how a on-prem environment would look like

3. And now hardware prices are going up because of AI

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Vision For The Future

- Being agnostic in where we run
  - Migrate our cluster away from managed clusters into VMs with CAPI
- PoC Talos + KubeVirt
- ARM CPU
  - Explorer ARM CPU for performance and cost on running our servers.
- GPU

Would We Suggest Everyone To Move?

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Thank you & QA

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